ScamShield US

- Agentic RAG for consumer fraud and identity-theft education
- Stack constraint: AWS EC2 and S3 only; open-source models; boto3
- Classify → retrieve top 5 → answer with local Qwen
- Generative AI & Natural Language Processing · Amir Jafari

Problem and Goals

- Help consumers interpret suspicious SMS messages safely
- Compare Traditional, Advanced (DistilBERT), and Hybrid classifiers
- Ground answers in official FTC, CISA, SSA, and related guidance

System Architecture

- Angular UI → FastAPI / Uvicorn API
- Agents: fraud detection → retrieval → Qwen answer → validation
- S3DataStore (boto3) for knowledge, models, indexes, and metrics

Offline Pipeline

- Seed knowledge → prepare splits → train models → build index
- Evaluate approaches → build dashboard visualizations
- All durable outputs written to the course S3 prefix

Online Investigation Workflow

- Hybrid classifier scores the message
- Hybrid BM25 and FAISS return five evidence chunks
- Qwen2.5-1.5B decoder generates a cited educational answer

Models From the Course Concept List

- Classical: TF-IDF with logistic regression / linear SVM
- Pretrained transformers: DistilBERT, MiniLM, Qwen
- Hybrid cascade margin = 0.05 (validation-chosen)

Results (Held-Out Test)

- Hybrid macro-F1 = 0.983; accuracy = 0.992; ~1.1% routed to DistilBERT
- Traditional and Advanced are strong; Hybrid wins on macro-F1
- Primary metric = macro-averaged F1 (rare subtypes matter)

Selected References

- Sanh et al. (2019). DistilBERT.
- Reimers & Gurevych (2019). Sentence-BERT / MiniLM embeddings.
- Johnson et al. (2019). FAISS / GPU similarity search.
- Qwen Team (2024). Qwen2.5 model card.
- Almeida et al. (n.d.); SMS Spam Research (n.d.). SMS data sets.

Demo and Reproducibility

- Code/ directory with uv sync and AWS credentials
- uvicorn backend.app.main:app --port 8000; npm start in frontend/
- Educational use only — not legal advice